

EE 345: AI assisted project

Fall 25
Due Fri Dec 19

We will look at something called Kalman Filters, a fundamental technique in engineering. It exists in every phone you have, and practically every GPS device you use. Every Global Navigation Satellite System (GNSS) chip from practically all chip manufacturers (Qualcomm, Broadcom, Sony, Mediatek, and anyone else you can think of) implement Kalman Filters in their firmware.

This assignment will proceed in phases. The entire assignment is due Dec 19 (the Fri of the finals week). There are soft deadlines to keep you on track. It is OK to miss them, but if you wish for feedback, please make the soft deadlines too.

When we understand a new topic, it is best to start with things you know.

1. (Due Mon Nov 24) Using the same projection arguments that we had for regular least squares, derive ℓ_2 -regularized least squares. Particularly, that if (using notation from class)

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \|\mathbf{y} - X\mathbf{w}\|^2 + \lambda \|\mathbf{w}\|^2$$

then

$$\hat{w} = (X^T X + \lambda I)^{-1} X^T \mathbf{y}.$$

Note that generic solutions tend to use calculus, but that is not very conceptually useful, hence we want projection arguments again. One could refine this further, instead of regularization using the length of $\mathbf{w}$, we could regularize using the quantity

$$\mathbf{w}^T A \mathbf{w}$$

for some matrix A . Now not all matrices A are meaningful in this context. Find out what matrices A would be inappropriate (in the sense that the optimization leads to meaningless solutions) and why.

2. (Due Fri Nov 28) We interpreted linear regression as projecting $\mathbf{y}$ into the column space of X , since $X\mathbf{w}$ is a linear combination of the columns of X . Another way to interpret linear regression (from a more traditional perspective, and one that makes more assumptions) is to think of $X\mathbf{w}$ to be linear measurements of $\mathbf{w}$. Therefore, our observation $\mathbf{y}$ is thought to be a noisy linear measurement of $X\mathbf{w}$, so we imagine that $\mathbf{y} = X\mathbf{w} + \mathbf{n}$, where $\mathbf{n}$ is a noise vector. Usually a probability model is assumed, but rather than worry about probability models, let us simply assume smaller lengths are more preferable.

We have been thinking of the vector $\mathbf{w}$ as something we find out from $\mathbf{y}$ and X , but otherwise completely arbitrary. What if $\mathbf{w}$ had some structure? For example if $\mathbf{w}$ had length $2n$, and you knew that the second n components of $\mathbf{w}$ had to be approximately the same as the first n components, ie it is periodic with period n (and that the first part had to be a vector with

as small a length as possible). Now if you were to find the best $\mathbf{w}$ using regular least squares, there is no guarantee that solution will enforce the above prior knowledge on $\mathbf{w}$. So we want to find a way to enforce this prior knowledge.

If you observed $\mathbf{y} = X\mathbf{w} + \mathbf{n}$, how would you enforce this constraint on $\mathbf{w}$ as ℓ_2 regularization?

Also extend to a general case where the components of $\mathbf{w}$ denote a time varying sequence, but the evolution of the sequence is linear. So $\mathbf{w}$ is a signal for us just written as a vector.

3. (Due Wed Dec 3) Now look up Kalman Filters. Observe that if you concatenated all the state vectors of the dynamical system, you get a vector with structure just like $\mathbf{w}$ in the prior part. Can you now write the Kalman filter as one large aggregated ℓ_2 regularized least squares problem?
4. (Due Mon Dec 8) *Observability* in Kalman filters refers to the ability to determine the initial state. From the previous part and from your understanding of when solutions exist, can you define observability? *Detectability* in Kalman filters means that even though you may not observe all of the initial state (ie you won't estimate it perfectly even if everything was noiseless), you can still be sure that the part of the state you did not observe will die out automatically in the dynamical evolution of the system. So what you do not observe will not hurt you. Express decidability again in terms of the spectral properties of the matrices involved in the Kalman filter.
5. (Due Fri Dec 19) Pick an application of Kalman filters and work it out in detail, explaining it using only the concepts above. I am not looking for a high level English language description. Rather, you have to do this part like an engineer, setting up the problem, variables, working it out fully, and giving me code (and simulated data to run it on).

Applications include heart beat tracking in health devices (your smart watches), tracking frequency in power systems, phase locked loops, or GPS systems.